HARIHARAN K

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EDUCATION

Chennai Institute of Technology, Chennai, India — Aug 2023 – May 2027
B.E. – Electronics and Communication Engineering — **CGPA: 8.89 / 10.0**

TECHNICAL SKILLS

- **Hardware & Embedded:** STM32 (HAL & Bare-Metal), ESP32, Arduino.
- **Programming Languages:** Embedded C, C, Java
- **Protocols & Middleware:** OpenDDS (Pub/Sub), CAN Bus, UART, SPI, I2C, BLE, MQTT
- **Sensors & Actuators:** MPU6050 (IMU), Ultrasonic, LDR Arrays, LM35, Servo Motors
- **Development Tools:** STM32CubeIDE, Keil μVision, LTSpice, Proteus
- **Core Concepts:** RTOS Fundamentals, Interrupt-Driven Architecture, Sensor Fusion, Closed-Loop Control Systems, PWM Actuation, ADC/GPIO, Digital Electronics, Network Theroy, Electronic Devices and Circutis.

PROFESSIONAL EXPERIENCE

Combat Vehicles Research and Development Establishment (CVRDE), DRDO — *Chennai, India*
Intern – Networking & Distributed Systems — Dec 2024 – Jan 2025

- Implemented **OpenDDS** publish-subscribe architectures on a 2-node distributed system, utilizing middleware paradigms foundational to modern robotics frameworks.
- Configured Reliability and Deadline QoS policies to minimize message delivery latency, ensuring real-time responsiveness.
- Analyzed middleware telemetry and networking protocols in the context of latency-sensitive defense applications.

Heavy Vehicles Factory (HVF) — *Chennai, India*
Intern – Manufacturing & Automation — May 2024 · 2 Weeks

- Analyzed CNC machining workflows for defense vehicle components, evaluating production line sequencing and mechanical fabrication efficiency.
- Configured and observed **PLC/SCADA** control systems during live industrial automation cycles on the factory floor.
- Evaluated automated quality inspection procedures and documented findings for heavy manufacturing systems.

PROJECTS

Embedded Anti-Collision & Autonomous Braking System — GitHub
Embedded C, STM32F4, MPU6050 (IMU), Ultrasonic, PWM Actuation

- Developed a safety-critical hardware prototype using **STM32F4** for real-time obstacle detection and automated response.
- Implemented **sensor fusion** by aggregating ultrasonic spatial data with MPU6050 accelerometer readings to filter noise and eliminate false-positive collision alerts.
- Engineered an interrupt-driven control pipeline achieving **sub-100 ms** end-to-end latency, instantly triggering PWM-based motor deceleration and GSM telemetry alerts.

2-DOF Closed-Loop Solar Tracker — GitHub
C, Microcontrollers, LDR Sensor Array, Servo Motors

- Designed a **2-Degree-of-Freedom (DOF)** closed-loop control system utilizing LDR arrays to autonomously track and orient toward peak light intensity.
- Implemented a differential feedback algorithm to drive precise **servo motor actuation**, achieving **~30–35% improvement** in energy capture over static systems.
- Optimized embedded power consumption by engineering low-power sleep state transitions during idle/low-light conditions.

Embedded Thermal Management & Alert System — GitHub
Embedded C, ESP32, LM35, ADC, PWM Control, I2C

- Developed an active cooling and thermal monitoring system for battery packs using **ESP32**, crucial for maintaining mobile hardware integrity.
- Programmed threshold-based control logic via real-time ADC acquisition to automatically actuate a **PWM-driven cooling fan** when temperatures exceed operational limits.
- Integrated an I2C OLED display and buzzer loop for continuous system status monitoring and failsafe alerts.

CERTIFICATIONS & ACHIEVEMENTS

- **IoT Challenge 2026** (FPT Software × Silicon Labs) – Shortlisted in **Top 50 Teams** as Team Lead for *NeuroGuard Edge*, developing an edge-computing wearable using **TinyML** on EFR32MG24 with BLE 5.3 telemetry.
- **NPTEL – IIT Delhi & Guwahati:** Transducers for Instrumentation *(Elite + Silver)*; Microsensors & Nanosensors *(Elite)*.
- **NPTEL:** Operating System Fundamentals *(Elite)*; Introduction to Industry 4.0 & Industrial IoT *(Elite)*.
- **Coursera:** Foundations of Project Management (Google); Introduction to IoT & Embedded Systems (UC Irvine).
- **Competitive Programming:** Solved **580+ problems** across LeetCode, Skillrack, and CodeChef.